Mini Project

Mini Block Cipher Algorithm

Master 1 cysia 2025-2026

Project Description:

This project aims to design and implement a **Mini Block Cipher** as an educational cryptographic system for learning the fundamental concepts of symmetric encryption. The cipher operates on small block and key sizes in order to keep the implementation simple, clear, and suitable for academic experimentation.

Objectives :

1. The main objective of this project is to understand how real-world block ciphers such as AES and DES are structured internally, including key scheduling, round functions, and diffusion and confusion mechanisms.
2. another key goal of this project is to **enrich the developer's professional profile** by building a high-quality, well-documented, and practical cryptography project that can be showcased on **GitHub** as part of a technical portfolio.

Task :

the task is to **design and implement the algorithm based on the specific parameters assigned to each students**, and to ensure that it correctly performs both **encryption and decryption**.

- **The competitive evaluation criteria among the students are:**

No.	Metric Name	Mathematical Symbol	Purpose
1	Shannon Entropy	$H(X)$	Measures the randomness and uncertainty of the ciphertext.
2	Correlation Coefficient	r_{xy}	Measures the statistical relationship between Plaintext and Ciphertext.
3	Strict Avalanche Criterion	SAC	Ensures that changing 1 bit in plaintext flips half of the ciphertext bits.
4	Hamming Distance	D_H	Measures the bit-level difference between original and encrypted text.
5	Frequency Test	S_n	Checks if the number of 0s and 1s in the ciphertext are approximately equal.
6	Time	-	Time (nano second) of 100 encryption & decryption

Term	Full Name	Explanation
Feistel	Feistel Network	A cipher structure where the block is split into two halves; a function is applied to one half and combined with the other using XOR. Decryption uses the same structure.
SPN	Substitution–Permutation Network	A structure based on repeated layers of S-Boxes (substitution) and P-Boxes (permutation) to achieve confusion and diffusion. Used in AES.
ROUNDS	Number of Rounds	The number of iterations of the cipher structure. More rounds usually mean higher security.
S-Box	Substitution Box	A nonlinear function that replaces values to provide confusion.
P-Box	Permutation Box	A function that rearranges bits or bytes to provide diffusion.
MIX_XOR	XOR Mixing	Mixing data using the XOR operation.
MIX_MDS	Maximum Distance Separable	A strong linear mixing matrix (like AES MixColumns) that provides optimal diffusion.
MIX_ADD_MOD	Modular Addition	Mixing using modular addition (usually mod 2^n).
MIX_NXOR	NOT-XOR Mixing	Mixing using XOR combined with bitwise NOT for extra nonlinearity.
ADD_KEY	Add Round Key	Combining the round key with the state using modular addition.
XOR_KEY	XOR with Key	Combining the round key with the state using XOR.
ROT_KEY	Rotate Key	Rotating the key bits left or right before using them.
Expansion	Key Expansion	Generating round keys from the main secret key (key schedule).
Permutation	Key Permutation	Reordering key bits to increase diffusion.

المصادر:

1. **TP 00: Python Basics for Cryptography**
2. **TP 02: Block Cipher**
3. **TD 02: block Cipher**
4. **DES**
5. **AES**
6. **TWOFISH**
7. **RC6**